RHC-UCRL

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1 Problem Statement

We assume a stochastic environment where the state space denoted as $\mathcal{S} \subseteq \mathbb{R}^p$ and the action space is denoted as $\mathcal{A} \subseteq \mathbb{R}^q$. The adversary actions are denoted as $\bar{\mathcal{A}} \subseteq \mathbb{R}^q$ At any state $s_h \in \mathcal{S}$ the transition dynamics is defined by function $f : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}}$. Both the action sets considered to be compact and the dynamics are given by

$$s_{h+1} = f(s_h, a_h, \bar{a}_h) + \omega_h \tag{1}$$

The reward and utility functions are given by $r : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to [0, 1]$ and $u : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to [0, 1]$ respectively.

Algorithm 1 RHC-UCRL

Input: intital state s_0, reward function $r : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to \mathbb{R}$, utility function $u : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to [0, 1]$, horizon length H, Iteration length τ,dual parameter λ initial policies $\pi_0, \bar{\pi}_0$ (Check if required)
for $t = 1, 2, \ldots, \tau$ **do**
    Reset the system to $s_{0,t+1} = s_0$
    for $h = 1, \ldots, H$ **do**
        $s_{(h,t)} = f(s_{(h-1,t)}, \pi_t(s_{(h-1,t)}), \bar{\pi}_t(s_{(h-1,t)})) + \omega_{h,t}$
        Collect $(s_{h-1}, a_{h-1}, \bar{a}_{h-1}, s_h, r, u)_t$
    end for
    Update the model dynamics to accurately express $f^{(o)}$ and $f^{(p)}$
    $\pi_t \in \arg\max_{\pi} \min_{\bar{\pi}} \left(J_r^{(o)} - \lambda \left[b - J_u^{(o)} \right]_+ \right)$
    $\bar{\pi}_t \in \arg\min_{\bar{\pi}} \left(J^{(p)}(\pi_t, \bar{\pi}) - \lambda \left[b - J_u^{(p)}(\pi_t, \bar{\pi}) \right]_+ \right)$
end for

Assumption 1. *The function f that determines the dynamics in equation* (1) *is L_f-Lipschitz continuous and the noise ω_h $\forall$ $h \in \{0, 1, \ldots, H-1\}$ are i.i.d. σ-Sub-Gaussian.*

Let us consider $J(f, \pi, \bar{\pi})$ and $J_u(f, \pi, \bar{\pi})$ as the expected long-run reward and the expected long-run utility value. These estimates evaluate the performance of a pair of polices $(\pi, \bar{\pi})$. The evaluations are performed on episodic data collected by playing the policy pair under evaluation. Formally, they can be defined as equation (2) and (3)

$$J(\tilde{f}, \pi, \bar{\pi}) := \mathbb{E}_{\nu_{\tilde{f},\pi,\bar{\pi}}} \left[\sum_{h=0}^{H} r(\mathbf{s}_h, \mathbf{a}_h, \bar{\mathbf{a}}_h) | \mathbf{s}_0 \right], \tag{2}$$

$$J_u(\tilde{f}, \pi, \bar{\pi}) := \mathbb{E}_{\nu_{\tilde{f},\pi,\bar{\pi}}} \left[\sum_{h=0}^{H-1} u(\mathbf{s}_h, \mathbf{a}_h, \bar{\mathbf{a}}_h) | \mathbf{s}_0 \right], \tag{3}$$

such that the state transition is given by $\mathbf{s_{h+1}} = \tilde{f}(\mathbf{s}_h, \mathbf{a}_h, \bar{\mathbf{a}}_h) + \omega_h$

The problem that we aim to solve is as mentioned in equation (4)

$$\begin{aligned} &\max_{\pi \in \Pi} \min_{\bar{\pi} \in \bar{\Pi}} J(f, \pi, \bar{\pi}), \\ &\textbf{s.t.} \ \min_{\bar{\pi} \leq \bar{\Pi}} J_u(s, \pi, \bar{\pi}) \geq b. \end{aligned} \tag{4}$$

Thus, we aim to find π^* defined as

$$\pi^* \in \arg\max_{\pi \in \Pi} \min_{\bar{\pi} \in \bar{\Pi}} \left[J(f, \pi, \bar{\pi}) - \lambda(b - J_u(f, \pi, \bar{\pi})) \right]. \tag{5}$$

To solve Algorithm 1, we need $J_t^{(o)}(\pi, \bar{\pi})$ and $J_t^{(p)}(\pi, \bar{\pi})$ defined in equations (6) and (7) respectively.

$$\begin{aligned} &J_t^{(o)}(\pi, \bar{\pi}) := \max_{\eta^{(o)}} J(f^{(o)}, \pi, \bar{\pi}), \\ &\text{s.t. } f^{(o)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) = \mu_{t-1}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) + \beta_t \eta^{(o)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) \Sigma_{t-1}^{1/2}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}). \end{aligned} \tag{6}$$

$$\begin{aligned} &J_t^{(p)}(\pi, \bar{\pi}) := \min_{\eta^{(o)}} J(f^{(p)}, \pi, \bar{\pi}), \\ &\text{s.t. } f^{(p)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) = \mu_{t-1}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) + \beta_t \eta^{(p)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) \Sigma_{t-1}^{1/2}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}). \end{aligned} \tag{7}$$

Performance metric: The aim is to find a policy π_T such that $\min_{\bar{\pi}} J(f, \pi_T, \bar{\pi}) \geq \min_{\bar{\pi}} J(f, \pi^*, \bar{\pi}) - \epsilon$ and $\min_{\bar{\pi}} J(f, \pi_T, \bar{\pi}) - b < \xi$

Statistical Model: We consider a model based RL approach. That is the goal is to learn the dynamics and find a near-optimal robust policy. We use statistical estimation to probabilistically reason about dynamic models $\tilde{f}$ that are compatible with the observed data $\mathcal{D}_{1:t} = \{\nu_{f,\pi_{t'},\bar{\pi}_{t'}}\}_{t'=1}^{t}$. This can be done by frequentist method or the bayesian perspective.

Assumption 2. *The statistical model is calibrated w.r.t f in* (1)*, so that with $\sigma_t(.) = diag(\Sigma_t(.))$ and a non-decreasing sequence of parameters $\{\beta_t\}_{t \geq 1} \in \mathbb{R}_+$, each depending on $\delta \in (0, 1)$, it holds jointly for all $t \geq 1$ and $\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}} \in \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}}$*

such that $|f(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})-\mu_{t-1}(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})| \leq \beta_t\sigma_{t-1}(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})$ *elementwise with probability at least* $1-\delta$

The calibrated model assumption (Assumption 2) is very common in literature [2]. This assumption is crucial for exploration: if the model is not well-calibrated, then leveraging its epistemic uncertainty does not provably guide exploration. For dynamics with bounded norm in a known RKHS, Assumption 2 is satisfied [6, 1]. In the case of neural network models, one-step-ahead predictions can be recalibrated [3].

We construct the set of plausible models at time t as

$$\mathcal{M}_t = \left\{ \tilde{f} \ : \ \left\| \tilde{f}(\cdot) - \mu_{t-1}(\cdot) \right\| \leq \beta_t \, \sigma_{t-1}(\cdot) \right\}.$$

By Assumption 2, we can guarantee that, with high probability, the true dynamics $f \in \mathcal{M}_t$ for all time steps t.

Assumption 3. *For every episode* t*, the functions* Σ_t*, the agent's policy* $\pi_t \in \Pi$*, the adversary's policy* $\bar{\pi}_t \in \bar{\Pi}$*, the reward function* $r(.,.,.)$ *and the utility function* $u(.,.,.)$ *are Lipschitz continuous with respective constants as* $L_\sigma, L_\pi, L_{\bar{\pi}}, L_r$ *and* L_u*.*

Assumption 4. *For all* $\mathbf{s} \in \mathcal{S}$*,* $\mathbf{a} \in \mathcal{A}$ *and* $\bar{\mathbf{a}} \in \bar{\mathcal{A}}$*, we assume that* $|r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})| \leq R_{max}$

Assumption 4 is a very common practice in RL literature. Mostly the reward is normalized to lie in $[0,1]$, which can be further scaled if required.

2 Proofs

Lemma 1. *(Adapted from Corollary 1 in [4]) Based on assumptions 1 and 3, for every* $\mathbf{s}, \mathbf{s}^{'} \in \mathcal{S}$ *the following ineqlity holds*

$$\|f(\mathbf{s},\pi(\mathbf{s}),\bar{\pi}(\mathbf{s})) - f(\mathbf{s}^{'},\pi(\mathbf{s}^{'}),\bar{\pi}(\mathbf{s}^{'}))\| \leq L_f\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\|\mathbf{s}-\mathbf{s}^{'}\|_2. \tag{8}$$

Proof.

$$\begin{aligned}
&\|f(\mathbf{s},\pi(\mathbf{s}),\bar{\pi}(\mathbf{s})) - f(\mathbf{s}',\pi(\mathbf{s}'),\bar{\pi}(\mathbf{s}'))\| \\
&\leq L_f\sqrt{\|\mathbf{s}-\mathbf{s}'\|_2^2 + \|\pi(\mathbf{s})-\pi(\mathbf{s}')\|_2^2 + \|\bar{\pi}(\mathbf{s})-\bar{\pi}(\mathbf{s}')\|_2^2} \\
&\leq L_f\sqrt{\|\mathbf{s}-\mathbf{s}'\|_2^2 + L_\pi^2\|\mathbf{s}-\mathbf{s}'\|_2^2 + L_{\bar{\pi}}^2\|\mathbf{s}-\mathbf{s}'\|_2^2} \\
&= L_f\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\,\|\mathbf{s}-\mathbf{s}'\|_2.
\end{aligned} \tag{9}$$

The first inequality holds due to the Lipschitz continuity of f. The second inequality is due to Lipschitz continuity of π and $\bar{\pi}$, which we assume in Assumptions 1 and 3. This completes the proof. □

Lemma 2. *(Adapted from Lemma 3 in [4]). Assuming the assumptions 1 and 3 are true, the following inequality holds:*

$$|J_r(f,\pi,\bar{\pi}) - J_r(\tilde{f},\pi,\bar{\pi})| \leq L_r\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|\right] \tag{10}$$

Proof.

$$\begin{aligned}
|J_r(f,\pi,\bar{\pi}) - J_r(\tilde{f},\pi,\bar{\pi})| &= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - \sum_{h=0}^{H-1} r(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} (r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - r(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\sum_{h=0}^{H-1}\mathbb{E}\left[(r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - r(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&\leq L_r\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|_2\right]
\end{aligned} \tag{11}$$

The first equality is through the definition of $J(f,\pi,\bar{\pi})$, the second equality is by converting the difference of total sum of utility following f and $\tilde{f}$. The third equality is obtained by converting the difference of the total utility into element wise difference as both runs for $H-1$ steps. The final inequality is due to Lipschitz property of utility, π and $\bar{\pi}$. This completes the proof. □

Lemma 3. *Assuming that assumptions 1 and 3 holds, the following inequality is true:*

$$|J_u(f,\pi,\bar{\pi}) - J_u(\tilde{f},\pi,\bar{\pi})| \leq L_u\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|\right] \tag{12}$$

Proof.

$$\begin{aligned}
|J_u(f,\pi,\bar{\pi}) - J_u(\tilde{f},\pi,\bar{\pi})| &= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} u(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - \sum_{h=0}^{H-1} u(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} (u(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - u(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\sum_{h=0}^{H-1}\mathbb{E}\left[(u(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - u(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&\leq L_u\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|_2\right]
\end{aligned} \tag{13}$$

The first equality is through the definition of $J_u(f, \pi, \bar{\pi})$, the second equality is by converting the difference of total sum of rewards following f and $\tilde{f}$. The third equality is obtained by converting the difference of the total rewards into element wise difference as both runs for $H-1$ steps. The final inequality is due to Lipschitz property of reward, π and $\bar{\pi}$. This completes the proof □

Lemma 4. *Assuming that assumptions 1, 2 and 3 are true, then from Lemma 3 following result holds*

$$|\min_{\bar{\pi}\in\bar{\Pi}} J_u(f, \pi, \bar{\pi}) - \min_{\bar{\pi}\in\bar{\Pi}} J_u(\tilde{f}, \pi, \bar{\pi})| \leq L_u\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \sum_{h=0}^{H-1} \mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|_2\right] \quad (14)$$

Proof.

$$\begin{aligned} |\min_{\bar{\pi}\in\bar{\Pi}} J_u(f, \pi, \bar{\pi}) - \min_{\bar{\pi}\in\bar{\Pi}} J_u(\tilde{f}, \pi, \bar{\pi})| &\leq \sup_{\bar{\pi}\in\bar{\Pi}} |J_u(f, \pi, \bar{\pi}) - J_u(\tilde{f}, \pi, \bar{\pi})| \\ &\leq L_u\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \sum_{h=0}^{H-1} \mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|_2\right] \end{aligned} \quad (15)$$

The first inequality is true since the difference between any two functions is always less than the supremum of the difference between the absolute values of the functions [5]. The second inequality follows from Lemma 3 and hence, the proof follows. □

Lemma 5. *Under assumptions 1, 2 and 3, for all episodes $t \geq 1$, any $\eta \in [-1, 1], h \in \{0, \ldots, H-1\}, \pi \in \Pi$ and $\bar{\pi} \in \bar{\Pi}$ the following inequality holds:*

$$\begin{aligned} \|\boldsymbol{s}_{h,t} - \tilde{\boldsymbol{s}}_{h,t}\| \leq 2\beta_{t-1} &\left((1 + L_f + 2\beta_{t-1}L_\sigma)\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}\right)^{h-1} \\ &\sum_{h'=0}^{h-1} \left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2. \end{aligned} \quad (16)$$

Proof. To avoid notational complexity, we use $L_{f,\pi} = L_f\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}$ and $L_{\sigma,\pi} = L_\sigma\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}$.

We prove by induction that

$$\|s_{h,t} - \tilde{s}_{h,t}\|_2 \leq 2.\beta_{t-1} \sum_{h'=0}^{h-1} (L_{f,\pi} + 2.\beta_{t-1}L_{\sigma,\pi})^{h-1-h'}.\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h',t})\| \quad (17)$$

For $h = 0$, clearly $s_{0,t} = \tilde{s_{0,t}}$. This is because we treat our initial state as fixed. We know, $s_{h+1,t} = f(s_{h,t}, \pi_t, \bar{\pi}_t)$. Thus, we can expand the LHS for $h+1$

as:

$$
\begin{aligned}
\|\mathbf{s}_{h+1,t} - \tilde{\mathbf{s}}_{h+1,t}\| &= \|f(\mathbf{s}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| \\
&= \|f(\mathbf{s}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) + f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| \\
&\leq \|f(\mathbf{s}_{h,t}, \pi_t, \bar{\pi}_t) - f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| + \|f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| \\
&\leq L_f \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \|\mathbf{s} - \tilde{\mathbf{s}}\|_2 + 2\beta_{t-1} \|\sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\tilde{\mathbf{s}}_{h,t})\|_2 \\
&= L_{f,\pi} \|\mathbf{s}_{h,t} - \tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1} \|\sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\tilde{\mathbf{s}}_{h,t}) + \sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t}) - \sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 \\
&\leq L_{f,\pi} \|\mathbf{s}_{h,t} - \tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1} (\|\sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\tilde{\mathbf{s}}_{h,t}) - \sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 + \|\sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t})\|_2) \\
&\leq L_{f,\pi} \|\mathbf{s}_{h,t} - \tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1} L_\sigma \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \|\mathbf{s}_{h,t} - \tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1} \|\sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 \\
&= (L_{f,\pi} + 2\beta_{t-1} L_{\sigma,\pi}) \|\mathbf{s}_{h,t} - \tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1} \|\sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 \\
&\leq 2\beta_{t-1} \sum_{h'=0}^{(h+1)-1} \left((L_f + 2\beta_{t-1} L_\sigma) \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \right)^{(h+1)-1-h'} \left\| \sigma_{t-1}^{\pi_t, \bar{\pi}_t}(\mathbf{s}_{h,t}) \right\|_2
\end{aligned}
\tag{18}
$$

Finally, notice that $(L_{f,\pi} + 2\beta_{t-1} L_{\sigma,\pi})^{(h-1-h')} < (1 + L_{f,\pi} + 2\beta_{t-1} L_{\sigma,\pi})^{h-1-h'} \leq (1 + L_{f,\pi} + 2\beta_{t-1} L_{\sigma,\pi})^{h-1}$ and the result follows when the above inequality is combined with final inequality from equation (18).

The first inequality is due to triangular inequality, the second inequality is a direct application of Lemma 1 and the second term comes from the assumption that both $f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) \in \mathcal{M}$ and $\tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) \in \mathcal{M}$. So $\|f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| = \|f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \mu(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) + \mu(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\|$. Now take the triangular inequality to get $\|f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| \leq \|f(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \mu(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| + \|\mu(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)\| = 2\beta_{t-1} \sigma_{t-1}(\tilde{\mathbf{s}}_{h,t}, \pi_t, \bar{\pi}_t)$.

The third inequality is by triangular inequality of $\|.\|_2$. The fourth inequality comes from Lipschitz property of $\sigma(.)$, π and $\bar{\pi}$. The final inequality is obtained by replacing inductive hypothesis with equation (17) □

Lemma 6. *Assuming assumption 4 holds, π_t is the chosen policy at instant t by Algorithm 1 and $c > 2$, the following inequality holds*

$$
[b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ \leq c R_{\max} / \lambda \tag{19}
$$

Proof.

$$
\begin{aligned}
&\lambda [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ \leq \min_{\bar{\pi}} J_r^{(o)}(\pi_t, \bar{\pi}) + \lambda [b - J_u^{(o)}(\pi_t, \bar{\pi})]_+ \\
&- \min_{\bar{\pi}} J_r^{(o)}(\pi^*, \bar{\pi}) - \lambda [b - J_u^{(o)}(\pi^*, \bar{\pi})]_+ + c R_{\max} \\
&\leq c R_{\max}
\end{aligned}
\tag{20}
$$

The first inequality is true because π^*, being the optimal policy, returns the highest value and the highest value cannot be more than R_{max}. Thus, $[b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ \leq c R_{\max} / \lambda$ □

Lemma 7. *Assuming assumptions 2 and 4 holds*

$$\alpha = \frac{cR_{\max}}{\lambda.2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]},$$

π^* be the optimal policy solving equation (4) and π_t and $\bar{\pi}_t$ be the policies selected by Algorithm 1 at instant t, then the following inequality holds

$$[b - J_u(f, \pi_t, \bar{\pi}_t)]_+ \leq 2L_u H\beta_T^H C^H (1+\alpha) \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]. \tag{21}$$

Proof.

$$\begin{aligned}
[b - J_u(f, \pi_t, \bar{\pi}_t)]_+ &= [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}) + \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}) - J_u(f, \pi_t, \bar{\pi}_t)]_+ \\
&\leq [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ + |\min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}_t) - J_u(f, \pi_t, \bar{\pi}_t)| \\
&\leq [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ + |J_u^{(o)}(\pi_t, \bar{\pi}_t) - J_u(f, \pi_t, \bar{\pi}_t)| \\
&\leq \frac{cR_{\max}}{\lambda} + L_u\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \sum_{h=0}^{H-1} \mathbb{E}[\|s_h - \tilde{s}_h\|_2] \\
&\leq \frac{cR_{\max}}{\lambda} + L_{u,\pi} \sum_{h=0}^{H-1} \mathbb{E}\Bigg[2\beta_{t-1}\Big((1 + L_f + 2\beta_{t-1}L_\sigma) \\
&\quad \cdot\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}\Big)^{h-1} \sum_{h'=0}^{h-1} \left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\Bigg] \\
&\leq \frac{cR_{\max}}{\lambda} + 2L_u\beta_T^H C^H \sum_{h=0}^{H-1} \mathbb{E}\left[\sum_{h'=0}^{h-1} \left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\right] \\
&\leq \frac{cR_{\max}}{\lambda} + 2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\right] \\
&= 2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\right] \\
&\left(\frac{cR_{\max}}{\lambda \cdot 2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\right]} + 1\right) \\
&= 2L_u H\beta_T^H C^H (1+\alpha) \sum_{h'=0}^{H-1} \mathbb{E}\left[\left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\right].
\end{aligned} \tag{22}$$

The first equality comes from adding and subtracting $\min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}_t)$. The first inequality is due to the triangle inequality of $||.||_\infty = \max a_1, a_2$, here $a_2 = 0$ and then invoking $[a]_+ = \max(a, 0) < |a|$. This inequality is true

because $a \leq |a|$ and $0 \leq |a|$. The second inequality is due to replacement of $\min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})$ with any adversarial policy $\bar{\pi}$. This is true since if we place the variable which returns the minimum functional value with any other variable in the candidate set, while the other function is fixed, the difference between the two values is bound to increase. Now, the third inequality is a direct application of Lemma 6 and Lemma 3. The fourth inequality is a direct application of Lemma 5. The fifth inequality is by assuming C^H which is the highest power of C as C^h was increasing term by term as C^0 when $h = 1$, C^1 when $h = 2$, thus, we replace all these C^{h-1} with the highest value of h which is H and take that common outside the summation. The sixth inequality is obtained by taking the outer summation $\sum_{h=0}^{H-1}$ inside expectation as expectation is a linear operator and then perform merging of the two summations into one by causing change of variables which spit out an additional $(H - h + 1)$ term inside the expectation. Now we consider this term $(H - h + 1)$ changes as h increases, so we replace each with the maximum value H and take that common. Next equality step is simply by taking the relevant terms common such that we can replace the second term with $(1 + \alpha)$ which follows from the definition of α above at the start of the lemma. □

Lemma 8. *Let π^* be the optimal feasible policy and the solution for equation 4 and let π_t and $\bar{\pi}_t$ be the polices selected by $RHC-UCRL$ at time t. Then under assumption 2 and $L_{(r,\lambda,u)} = (L_r + \lambda L_u)$, the following inequality (equation (23)) hold with probability at least $1 - \delta_r$.*

$$\min_{\bar{\pi}} J_r(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}} J_r(f, \pi_t, \bar{\pi}) \leq 4L_{(r,\lambda,u)} \beta_T^H C^H H \sum_{h'=0}^{h-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]. \tag{23}$$

Proof. Let us consider $\min_{\bar{\pi}} J(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}} J(f, \pi_t, \bar{\pi})$ as the instantaneous robust regret of the selected policy π_t and the quantity $\min_{\bar{\pi}} J_u(f, \pi_t, \bar{\pi}) - b$ as the instantaneous robust violation of the same policy. We first determine the

instantaneous regret bound, followed by the instantaneous violation bound.

$$
\begin{aligned}
&\min_{\bar{\pi}} J_r(f,\pi^*,\bar{\pi}) - \min_{\bar{\pi}} J_r(f,\pi_t,\bar{\pi}) \leq \min_{\bar{\pi}} J^{(o)}_{r_t}(\pi^*,\bar{\pi}) - \min_{\bar{\pi}} J(f,\pi_t,\bar{\pi}) \\
&\leq \min_{\bar{\pi}} \left(J^{(o)}_{r_t}(\pi^*,\bar{\pi}) + \lambda[b - J^{(o)}_u(\pi^*,\bar{\pi})]_+ \right) - \min_{\bar{\pi}} J_r(f,\pi_t,\bar{\pi}) \\
&\leq \min_{\bar{\pi}} \left(J^{(o)}_{r_t}(\pi_t,\bar{\pi}) + \lambda[b - J^{(o)}_u(\pi_t,\bar{\pi})]_+ \right) - \min_{\bar{\pi}} J_r(f,\pi_t,\bar{\pi}) \\
&\leq J^{(o)}_{r_t}(\pi_t,\bar{\pi}_t) + \lambda[b - J^{(o)}_u(\pi_t,\bar{\pi}_t)]_+ - \min_{\bar{\pi}} \left(J_r(f,\pi_t,\bar{\pi}) + \lambda[b - J_u(\pi_t,\bar{\pi})]_+ \right) \\
&\leq J^{(o)}_{r_t}(\pi_t,\bar{\pi}_t) + \lambda[b - J^{(o)}_u(\pi_t,\bar{\pi}_t)]_+ - \min_{\bar{\pi}} \left(J^{(p)}_r(f,\pi_t,\bar{\pi}) + \lambda[b - J^{(p)}_u(\pi_t,\bar{\pi}_t)]_+ \right) \\
&\leq J^{(o)}_{r_t}(\pi_t,\bar{\pi}_t) + \lambda[b - J^{(o)}_u(\pi_t,\bar{\pi}_t)]_+ - J^{(p)}_r(\pi_t,\bar{\pi}_t) - \lambda[b - J^{(p)}_u(\pi_t,\bar{\pi})]_+ \\
&\leq |J^{(o)}_{r_t}(\pi_t,\bar{\pi}_t) - J_r(f,\pi_t,\bar{\pi}_t)| + |J^{(p)}_r(\pi_t,\bar{\pi}) - J_r(f,\pi_t,\bar{\pi}_t)| + \lambda[J^{(p)}_u(\pi_t,\bar{\pi}_t) - J^{(o)}_u(\pi_t,\bar{\pi}_t)]_+ \\
&\leq |J^{(o)}_{r_t}(\pi_t,\bar{\pi}_t) - J_r(f,\pi_t,\bar{\pi}_t)| + |J^{(p)}_r(\pi_t,\bar{\pi}) - J_r(f,\pi_t,\bar{\pi}_t)| + \lambda|J^{(p)}_u(\pi_t,\bar{\pi}_t) - J^{(o)}_u(\pi_t,\bar{\pi}_t)| \\
&\leq |J^{(o)}_{r_t}(\pi_t,\bar{\pi}_t) - J_r(f,\pi_t,\bar{\pi}_t)| + |J^{(p)}_r(\pi_t,\bar{\pi}) - J_r(f,\pi_t,\bar{\pi}_t)| \\
&+ \lambda|J^{(p)}_u(\pi_t,\bar{\pi}_t) - J_u(f,\pi_t,\bar{\pi}_t)| + |J^{(o)}_u(\pi_t,\bar{\pi}_t) - J_u(f,\pi_t,\bar{\pi}_t)| \\
&\leq L_{r,\pi} \sum_{h=0}^{H} \mathbb{E}\left[\|\mathbf{s}_{h,t} - \mathbf{s}^{(o)}_{h,t}\|_2 + \|\mathbf{s}_{h,t} - \mathbf{s}^{(p)}_{h,t}\|_2 \right] + \lambda L_{u,\pi} \sum_{h=0}^{H} \mathbb{E}\left[\|\mathbf{s}_{h,t} - \mathbf{s}^{(o)}_{h,t}\|_2 + \|\mathbf{s}_{h,t} - \mathbf{s}^{(p)}_{h,t}\|_2 \right]
\end{aligned}
\tag{24}
$$

The first inequality comes from converting the the model function f with the optimistic model function $f^{(o)}$ which is sure to return the maximum expected return for π^*. The second inequality is true due to addition of an extra positive term to RHS. The third inequality comes by rplacing the optimal policy by the protagonistic policy returned at time t. The fourth inequality is true since we remove the minimum $\bar{\pi}$ with any other policy π_t whose return value is certain to be above the one returned by $\min_{\bar{\pi}}(J^{(o)}_{r_t}(.,\bar{\pi}) + \lambda[b - J^{(o)}_u(.,\bar{\pi})]_+)$. The fifth optimality is obtained by transitioning the model function f in the negative term with a lower returning pessimistic path $f^{(p)}$. Now basically for the second term in the sixth inequality, we have out solution $\bar{\pi}_t$ returned by `RHC-UCRL`. Hence, we replace that. Now, in the seventh inequality, we combine the different terms, take their absolute values, and impose the inequality $x \leq |x|$. In the eigth inequality, we replace the $[.]_+$ with $|.|$ which follows the relation $[x]_+ \leq |x|$. The ninth inequality is an application of triangle inequality and in the tenth inequality we directly apply Lemma 2 and Lemma 3.

Then upon continuation of equation (24), we use Lemma 5, it follows that all the terms inside the expectation are bounded in the same way as follows:

$$
\begin{aligned}
\|\mathbf{s}_{h,t} - \mathbf{s}^{(o)}_{h,t}\|_2 \leq 2\beta_{t-1} &\left((1 + L_f + 2\beta_{t-1} L_\sigma) \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \right)^{h-1} \\
&\sum_{h'=0}^{h-1} \left\| \sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big) \right\|_2 ,
\end{aligned}
\tag{25}
$$

as $f^{(o)}$ and $f^{(p)}$ belong to the same plausible models $\mathcal{M}_t$. By applying the above inequality twice and denoting $C := \left((1 + L_f + 2L_\sigma)\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}\right)$, we arrive at

$$\begin{aligned}
\min_{\bar{\pi}} J_r(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}} J_r(f, \pi_t, \bar{\pi}) &\leq 4(L_r + \lambda L_u)\beta_T^H C^H \\
&\sum_{h=0}^{H} \mathbb{E}\left[\sum_{h'=0}^{h-1} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right], \\
&= 4L_{(r,\lambda,u)}\beta_T^H C^H \sum_{h=0}^{H} \mathbb{E}\left[\sum_{h'=0}^{h-1} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right] \\
&\leq 4L_{(r,\lambda,u)}\beta_T^H C^H H \sum_{h'=0}^{h-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]
\end{aligned} \tag{26}$$

where $t \leq T$ and $1 \leq \beta_t$ is non-decreasing in t. The last inequality is obtained by taking the outer summation $\sum_{h=0}^{H-1}$ inside the expectation as expectation is a linear operator, and then performing the merging of the two summations into one by causing a change of variables, which results in an additional $(H - h + 1)$ term inside the expectation. Now we consider the term $(H - h + 1)$; as h increases, it changes, so we replace each occurrence with the maximum value H and take that common.

□

Theorem 1. *Under Assumptions 1 to 3 and considering*

$$C = \left((1 + L_f + 2L_\sigma)\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}\right),$$

$s_{h,t} \in \mathcal{S}$, $a_{h,t} \in \mathcal{A}$ *and* $\bar{a}_{h,t} \in \bar{\mathcal{A}}$ *for all* $t, h > 0$ *and* $\sigma_{t-1}^h = \sigma_{t-1}(s_{h,t}, \pi_t(s_{h,t}), \bar{\pi}(s_{h,t}))$. *Then for a fixed* $H \geq 1$, *with probability at least* $1 - \delta_r$, *the robust cumulative regret of RHC-UCRL is upper bounded by:*

$$R_T = \mathcal{O}\left(L(r, \lambda, u)\beta_T^H C^H H^{3/2}\sqrt{T\Gamma_T}\right),$$

and with probability at least $1 - \delta_u$,

$$V_T = \mathcal{O}\left(L_u \beta_T^H C^H (1 + \alpha) H^{3/2}\sqrt{T\Gamma_T}\right).$$

Proof. We first bound the cumulative regret and then the cumulative violation.

$$R_T = \sum_{t=1}^{T} \underbrace{\min_{\bar{\pi}\in\bar{\Pi}} J(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}\in\bar{\Pi}} J(f, \pi_t, \bar{\pi})}_{:= r_t} \tag{27}$$

$$\leq \sqrt{T\sum_{t=0}^{T} r_t^2} \tag{28}$$

$$\leq 4L_{(r,\lambda,u)}\beta_T^H C^H H\sqrt{T}\sqrt{\sum_{t=1}^{T}\left(\mathbb{E}\left[\sum_{h'=0}^{H-1}\left\|\sigma_{t-1}^{h'}\right\|_2\right]\right)^2} \tag{29}$$

$$\leq 4L_{(r,\lambda,u)}\beta_T^H C^H H\sqrt{T}\sqrt{\sum_{t=1}^{T}\mathbb{E}\left(\left[\sum_{h'=0}^{H}\left\|\sigma_{t-1}^{h'}\right\|_2\right]^2\right)} \tag{30}$$

$$\leq 4L_{(r,\lambda,u)}\beta_T^H C^H H^{1.5}\sqrt{T}\sqrt{\underbrace{\sum_{t=1}^{T}\mathbb{E}\left(\left[\sum_{h'=0}^{H}\left\|\sigma_{t-1}^{h'}\right\|_2^2\right]\right)}_{\Gamma_T}}. \tag{31}$$

where inequality (28) comes from Cauchy-Schwarz inequality; equation (29) is directly taken from our Lemma 8 ; equation (30) is true due to application of Jensen's inequality; equation (31) is true due to application of Cauchy-Schwarz's inequality to equation (30) and then by definition of Γ_T, and the required proof follows

Similarly, we now derive the same for the cumulative Violation bound:

$$V_T = \sum_{t=1}^{T}\underbrace{\big(b - J(f,\pi_t,\bar{\pi}_t)\big)_+}_{v_t} \tag{32}$$

$$\leq \sqrt{T\sum_{t=1}^{T}v_t^2} \tag{33}$$

$$\leq 2L_u H\beta_T^H C^H(1+\alpha)\sqrt{T\sum_{t=1}^{T}\left(\sum_{h=0}^{H}\mathbb{E}\left[\sum_{h'=0}^{H}\left\|\sigma_{t-1}^{h'}\right\|_2\right]\right)^2} \tag{34}$$

$$\leq 2L_u H\beta_T^H C^H(1+\alpha)\sqrt{T\sum_{t=1}^{T}\mathbb{E}\left(\left[\sum_{h'=0}^{H}\left\|\sigma_{t-1}^{h'}\right\|_2\right]^2\right)} \tag{35}$$

$$\leq 2L_u\beta_T^H C^H(1+\alpha)H^{1.5}\sqrt{T}\sqrt{\underbrace{\sum_{t=1}^{T}\mathbb{E}\left(\left[\sum_{h'=0}^{H}\left\|\sigma_{t-1}^{h'}\right\|_2^2\right]\right)}_{\Gamma_T}} \tag{36}$$

where the first inequality is true due to the Cauchy-Schwarz inequality; the second inequality is due to Lemma 7, the third inequality is due to application of Jensen's inequality and the final inequality is due application of Cauchy-Schwarz's inequality. Finally, we use the definition of Γ_T and the statement follows. This completes the required proof. □

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